

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without permission of BIS or used as an Indian Standard)

भारतीय मानक मसौदा
छेनी दांत संरक्षित (बड़े पशुओं) — विशिष्टि

(IS 12302 का पहला पुनरीक्षण)

Draft Indian Standard

Chisel Tooth Guarded (Large Animals) — Specification

(First Revision of IS 12302)

ICS 11.220

Veterinary Hospital Planning and Surgical
Instruments Sectional Committee, MHD 13

Last date for comments: **XX September 2026**

FOREWORD

(Formal clauses will be added later)

This Indian Standard was first published in 1988 as ‘Specification for Chisel Tooth Guarded (Large Animals)’. The first revision of this standard has been brought out to align the cross references to the latest standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (second revision)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This Indian Standard covers the requirements for chisel, tooth, guarded used in veterinary surgery for large animals.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

IS No.	Title
IS 1068:2025	Metallic and Other Inorganic Coatings — Electrodeposited Coatings of Nickel, Nickel Plus Chromium, Copper Plus Nickel and of Copper Plus Nickel Plus Chromium (<i>fourth revision</i>)
IS 2102 (Part 1) : 1993	General Tolerances Part 1 Tolerances for Linear and Angular Dimensions Without Individual Tolerance Indications (<i>third revision</i>)
IS 2500 (Part 1) : 2000	Sampling procedures for inspection by attributes: Part 1 sampling schemes indexed by acceptance quality limit (AQL) for lot - By - Lot inspection (<i>third revision</i>)
IS 3748 : 2022	Tool Steels — Specification (<i>third revision</i>)

3 MATERIAL

Shall conform to designation C120U of IS 3748.

4 SHAPE AND DIMENSIONS

4.1 Shall Conform to Fig. 1.

4.2 The tolerance on dimensions of working ends shall conform to ‘fine’ class of deviation and the tolerance for remaining dimensions shall conform to ‘medium’ class of deviation of IS 2102 (Part 1).

5 MANUFACTURE WORKMANSHIP AND FINISH

5.1 The chisel shall be manufactured in one piece. The central line running through the axis of the handle shall coincide with the base of the blade. The edges shall be even except for the cutting edge which shall be rounded. The cutting edge shall be sharp.

5.2 The mass of the chisel shall be between 450 to 550 g.

5.3 The surface of the working ends shall be free from all defects like burrs, pits, cracks, feathers, nicks and waviness when examined under X 10 magnification. All surfaces shall be free from pores, crevices and grinding marks.

5.4 The instrument shall be supplied free from residual scales, acid, grease and grinding and polishing materials. Compliance with these requirements shall be checked under X 2 magnification.

6 HARDNESS

The chisel shall be hardened and tempered to achieve the following hardness:

- a) Blade — 650 to 700 *HV*, and
- b) Handle — 400 to 500 *HV*.

7 REQUIREMENTS

The chisel shall be plated chromium over nickel conforming to service condition number 2 of IS 1068, except bevelled edge which shall be sharpened.

8 TESTS

8.1 Performance Test

The cutting edge of the blade of the chisel shall have a piece of hardwood such as *Kusum* or *Bulletwood* quickly and cleanly. On completion of the test, the chisel shall not show any sign of damage.

8.2 Load Test

The handle of the chisel at a point approximately 25 mm from the proximal end shall be fixed in a vice (with the centre line parallel with the jaws of the vice). A proof load of 34 kg, *Min* shall be applied at the proximal end of the handle. On completion of the test, the chisel shall not take a permanent set.

8.3 Corrosion Test

8.3.1 The instrument shall be cleaned with a suitable organic solvent (for example, carbon tetrachloride) in order to remove any trace of grease before carrying out the test.

8.3.2 The instrument shall be boiled for 60 minutes in 5 percent aqueous solution of sodium carbonate. At the end of this period, the plating shall not show any sign of peeling off nor shall there be any rusting over the plated portion.

8.3.3 The instrument shall be placed on a perforated platform on a desiccator above a solution of sodium chloride (20 gm NaCl in 100 cc of water of specific gravity of 1.50). Air shall be blown into the solution so that a dense fog is produced in the desiccator without being blown directly against the instrument. There shall be no sign of corrosion of the plating after treating for 24 hours.

9 SAMPLING

9.1 Unless otherwise agreed to between the purchaser and the supplier, the sampling procedure given in IS 2500 (Part 1) shall be followed.

9.2 Lot

In a single consignment, all the chisels of same size, shape and belonging to the same batch of manufacture shall be grouped together to constitute a lot.

9.3 For dimensions and requirements given in 4 and 5 a single sampling plan with general inspection level II and AQL 4 percent given in Tables 1 and 2-A of IS 2500 (Part 1) shall be followed.

9.4 For the requirement given in 7, a single sampling plan with general inspection level I and AQL 4 percent given in Tables 1 and 2-A of IS 2500 (Part 1) shall be followed.

9.5 For the tests given in 6 and 8, a single sampling plan with special inspection level S-3 and AQL 0.40 percent given in Tables 1 and 2-A of IS 2500 (Part 1) shall be followed.

10 MARKING

10.1 Each chisel shall be legibly and indelibly marked with the manufacturer's name, initials or recognized trade-mark.

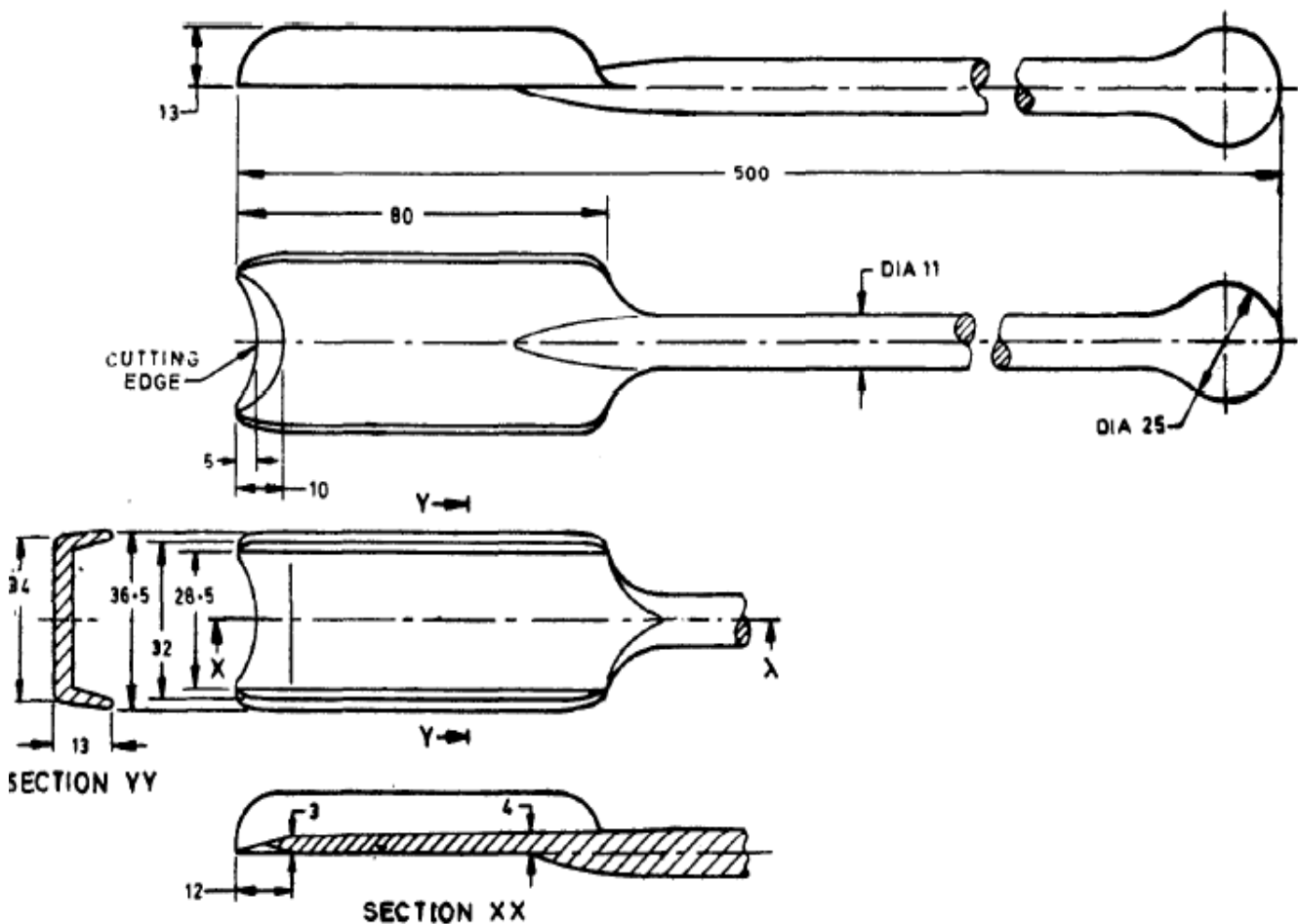
10.2 BIS Certification Marking

The product(s) confirming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

11 PACKING

11.1 The chisel shall be delivered in a new and cleaned condition free from all traces of rust.

11.2 Each chisel shall be coated with thin film of protective coatings and -wrapped in suitable wax paper. The empty space, if any, shall be filled with suitable cushioning material to prevent movement.



All dimensions in millimetres.

FIG. 1 CHISEL TOOTH GUARDED